Running your own Word Count Program

Before completing the below tasks make sure that you have :

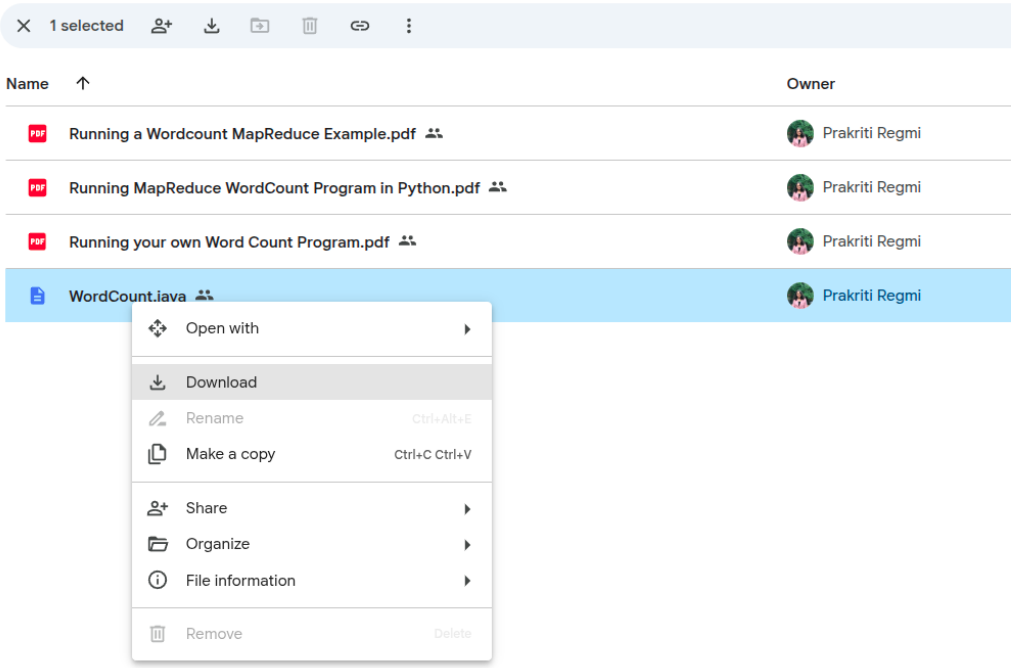
- Switched to the **hadoop user**.
- Navigated to the hadoop **Sbin** directory.
- Started the **Hadoop Distributed File System (HDFS).**
- Started the **YARN Resource Manager**.

If not refer to the below image to complete the above tasks :

```
hdoop@Siddhartha-Shakya: ~
siddhu@Siddhartha-Shakya:~$ jps
441 Jps
siddhu@Siddhartha-Shakya:~$ su - hdoop
Password:
hdoop@Siddhartha-Shakya:~$ cd hadoop-3.2.4/sbin/
hdoop@Siddhartha-Shakya:~/hadoop-3.2.4/sbin$ ./start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Siddhartha-Shakya]
hdoop@Siddhartha-Shakya:~/hadoop-3.2.4/sbin$ ./start-yarn.sh
Starting resourcemanager
Starting nodemanagers
hdoop@Siddhartha-Shakya:~/hadoop-3.2.4/sbin$ jps
2020 Jps
1477 ResourceManager
1626 NodeManager
781 NameNode
910 DataNode
1183 SecondaryNameNode
hdoop@Siddhartha-Shakya:~/hadoop-3.2.4/sbin$
```

If **jps (Java Processing Status)** lists all the above **running java processes**, then you are **good to go.**

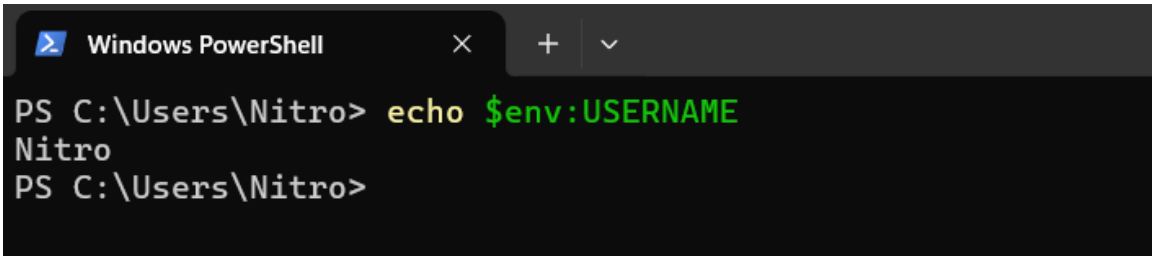
1. Download **WordCount.java** file from the drive.



Note : Make sure that the **java file** you downloaded is in **Downloads.**

2. In **windows powershell** type in the following command.

```
> echo $env:USERNAME
```



Take note of the output, **mine is Nitro** yours **could be different**.

3. Move your **WordCount.java** file to your **WSL home directory**.

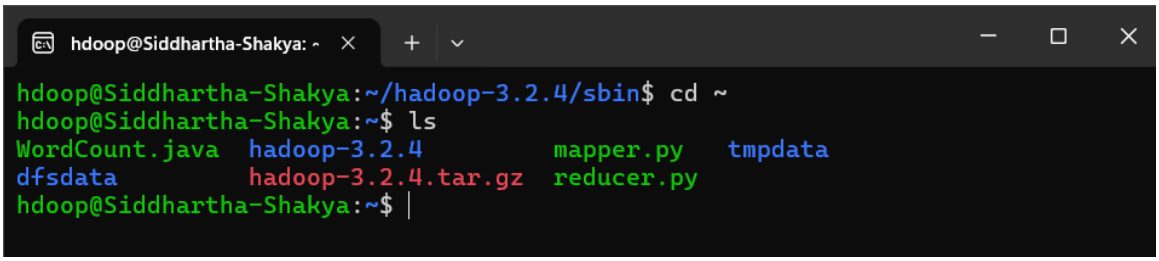
```
~$ cp  /mnt/c/Users/YourUser/Downloads/WordCount.java ~/
```

Important: Don't forget to replace **"YourUser"** with the **output from the above step 2**.

4. Check if the **WordCount.java file** has been **successfully copied** to your **WSL home directory.**

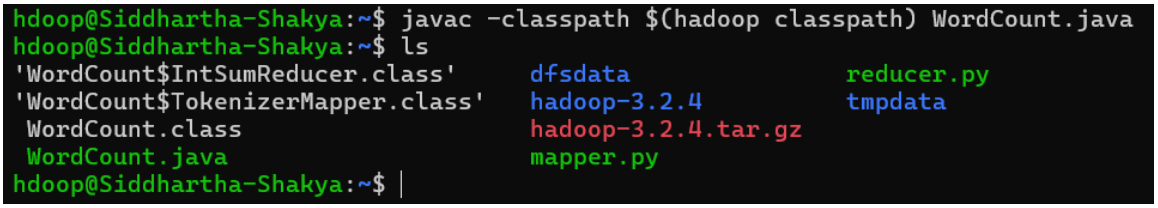
```
~$ cd ~

~$ ls
```



5. Compile the file :

```
~$ javac -classpath $(hadoop classpath) WordCount.java
```



6. Produce the Jar file :

```
~$ jar cf wordcount.jar Word*.class
```

7. Create the input directory on the hdfs :

```
~$ hdfs dfs -mkdir /input_word

~$ hdfs dfs -ls /
```

8. Create the input files :

```
~$ echo A long time ago in a galaxy far far away > testfile1

~$ echo Another episode of Star Wars > testfile2
```

9. Save the files to the input directory :

```
~$ hdfs dfs -put testfile? /input_word
```

10. Verify the uploads by checking if the files are inside the input_word directory.

```
~$ hdfs dfs -ls /input_word
```

11. It is important that the output directory does not already exist. If you have run the program before you need to delete the previous output directory :

```
~$ hdfs dfs -rm -R /output_word
```

12. Run the Map Reduce program :

```
~$ hadoop jar wordcount.jar WordCount /input_word /output_word
```

13. Check what files are in the output directory :

```
~$ hdfs dfs -ls /output_word
```

14. See what is in the output file:

```
~$ hdfs dfs -cat /output_word/part-r-00000
```

15. If you want to use the output file outside the hdfs you have to retrieve it:

```
~$ hdfs dfs -get /output_word/part-r-00000 word-results.txt
```